

EXPERIMENT 13:

The screenshot shows a VS Code editor with a project named 'FSD-2'. The Explorer panel on the left shows the project structure, including a 'venv' directory and an 'app.py' file. The main editor window displays the code for 'app.py', which includes a 'delete_student' function and a Flask app instance. The bottom panel shows the 'TERMINAL' output, which contains the command to run the application and its output, including a warning about using a development server and the debug mode status.

```
Experiment-13 > app.py > ...
66 def delete_student(id):
67     db.session.commit()
68     return jsonify({"message": "Deleted"})
69
70 if __name__ == '__main__':
71     app.run(debug=True)
```

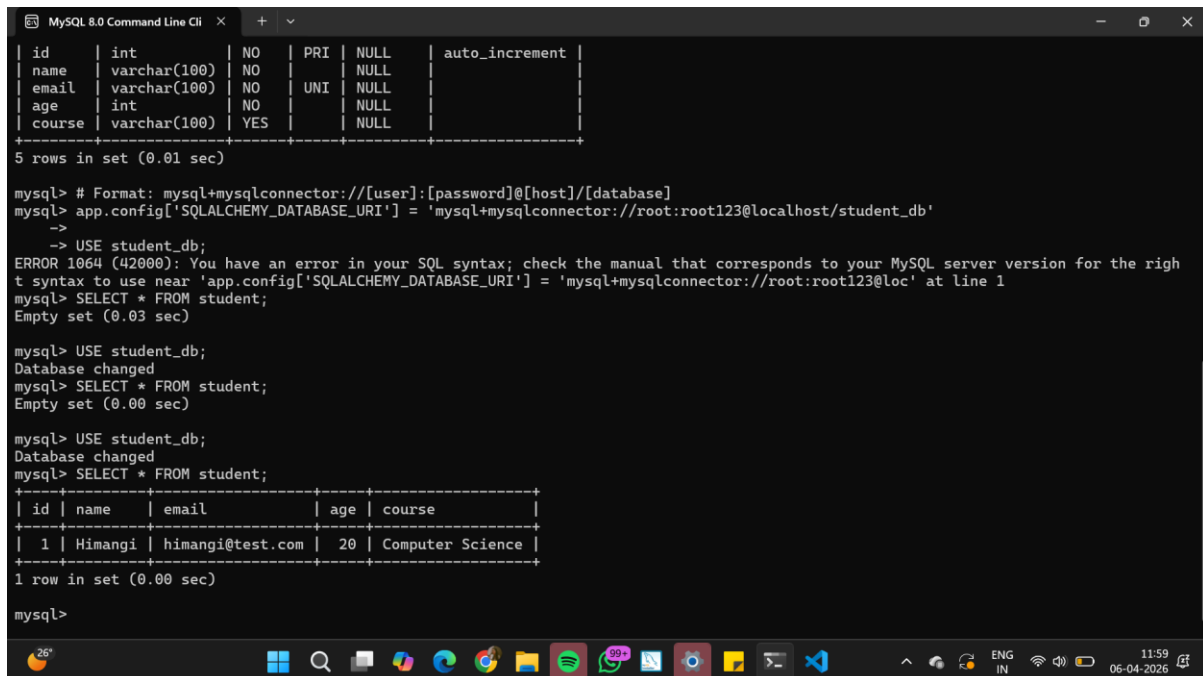
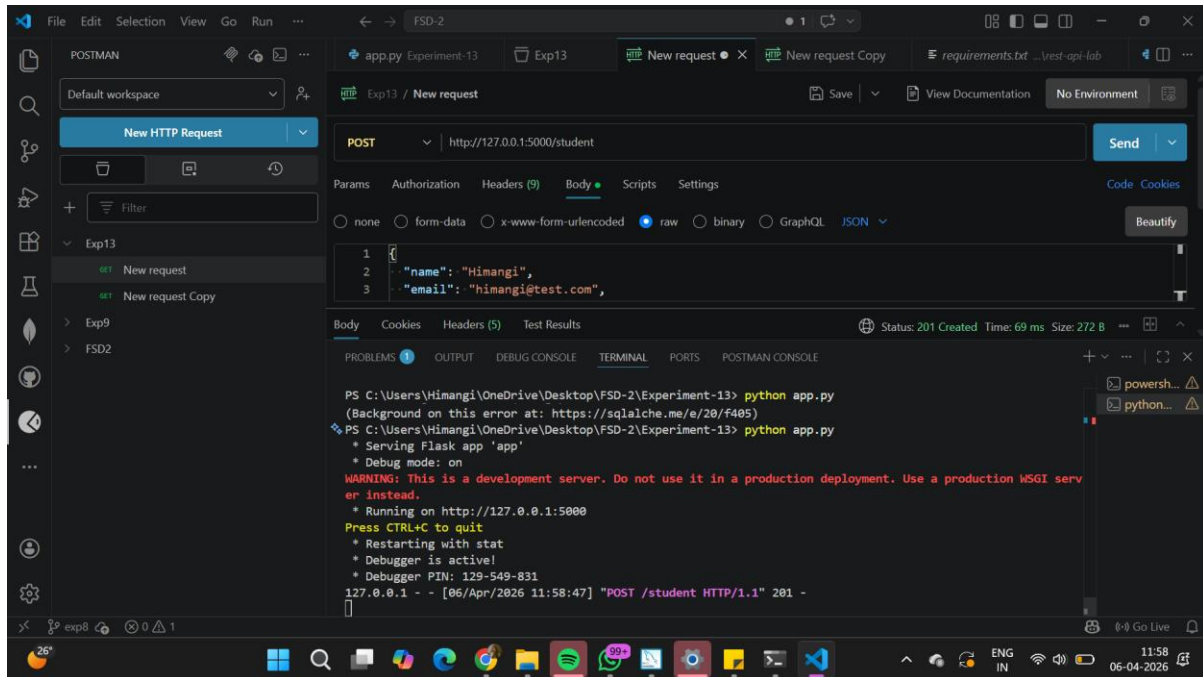
PS C:\Users\Himangi\OneDrive\Desktop\FSD-2> cd Experiment-13
PS C:\Users\Himangi\OneDrive\Desktop\FSD-2\Experiment-13> python app.py
C:\Users\Himangi\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.12_qbz5n2kfra8p0\python.exe: can't open file 'C:\Users\Himangi\OneDrive\Desktop\FSD-2\Experiment-13\app.py': [Errno 2] No such file or directory
PS C:\Users\Himangi\OneDrive\Desktop\FSD-2\Experiment-13> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 129-549-831

mysql> -- 2. Switch to the database
mysql> USE student_db;
Database changed
mysql>
mysql> -- 3. Create the table
mysql> CREATE TABLE student (
--> id INT AUTO INCREMENT PRIMARY KEY,
--> name VARCHAR(100) NOT NULL,
--> email VARCHAR(100) UNIQUE NOT NULL,
--> age INT NOT NULL,
--> course VARCHAR(100)
-->);
Query OK, 0 rows affected (0.08 sec)

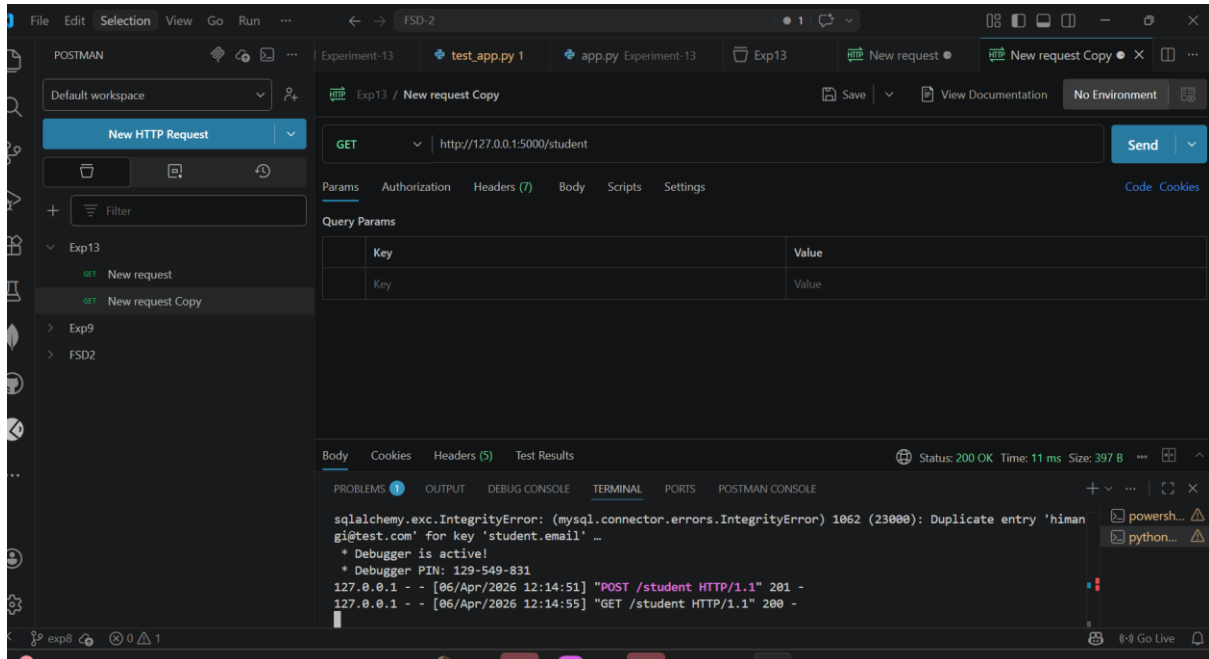
mysql>
mysql> -- 4. Verify the table exists (Optional)
mysql> DESC student;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
id	int	NO	PRI	NULL	auto_increment
name	varchar(100)	NO		NULL	
email	varchar(100)	NO	UNI	NULL	
age	int	NO		NULL	
course	varchar(100)	YES		NULL	
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.01 sec)

mysql> # Format: mysql+mysqlconnector://[user]:[password]@[host]/[database]
mysql> app.config['SQLALCHEMY_DATABASE_URI'] = 'mysql+mysqlconnector://root:root123@localhost/student_db'
--> |

CREATE STUDENT:



READ STUDENT:



1 row in set (0.00 sec)

```
mysql> USE student_db;
```

Database changed

```
mysql> SELECT * FROM student;
```

id	name	email	age	course
1	Himangi	himangi@test.com	20	Computer Science
4	Himangi	himangi2@test.com	20	Computer Science

2 rows in set (0.00 sec)

```
mysql> |
```

UPDATE STUDENT:

The screenshot shows the Postman interface for a PUT request. The URL is `http://127.0.0.1:5000/student/1`. The request body is a JSON object: `{ "course": "Data Science" }`. The response status is `200 OK` with a time of `33 ms` and size of `262 B`. The response body is a JSON object: `{ "age": 20, "course": "Data Science", "email": "himangi@test.com", "name": "Himangi" }`.

```
PUT http://127.0.0.1:5000/student/1
```

```
{
  "course": "Data Science"
}
```

```
{
  "age": 20,
  "course": "Data Science",
  "email": "himangi@test.com",
  "name": "Himangi"
}
```

DELETE STUDENT:

The screenshot shows the Postman interface for a DELETE request. The URL is `http://127.0.0.1:5000/student/1`. The response status is `200 OK` with a time of `16 ms` and size of `193 B`. The response body is a JSON object: `{ "message": "Deleted" }`.

```
DELETE http://127.0.0.1:5000/student/1
```

```
{
  "message": "Deleted"
}
```

2 rows in set (0.00 sec)

```
mysql> SELECT * FROM student;
```

id	name	email	age	course
4	Himangi	himangi2@test.com	20	Computer Science

1 row in set (0.01 sec)

VALIDATION CHECK:

The image shows the Postman application interface. At the top, a POST request is configured to `http://127.0.0.1:5000/student`. The 'Body' tab is selected, and the request body is a JSON object: `{ "name": "Himangi", "email": "himangi@test.com", "age": 10, "course": "Computer Science" }`. The 'raw' format is selected for the body. Below the body editor, the status bar indicates a **400 Bad Request** with a time of 48 ms and a size of 267 B. At the bottom, the terminal shows the command `python app.py` being executed in a PowerShell window, and the output shows the POST request failing with a 400 status code.

```
POST http://127.0.0.1:5000/student
```

Params Authorization Headers (9) **Body** Scripts Settings Code Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON** Beautify

```
1 {
2   "name": "Himangi",
3   "email": "himangi@test.com",
4   "age": 10,
5   "course": "Computer Science"
6 }
```

Body Cookies Headers (5) Test Results Status: 400 Bad Request Time: 48 ms Size: 267 B

PROBLEMS (1) OUTPUT DEBUG CONSOLE **TERMINAL** PORTS POSTMAN CONSOLE

```
PS C:\Users\Himangi\OneDrive\Desktop\FSD-2\Experiment-13> python app.py ...
127.0.0.1 - - [06/Apr/2026 12:00:00] "POST /student HTTP/1.1" 400 -
```

powersh... python...